

Chapter 3

Benefits, Costs and Decisions

Learning Objectives

- 3.1 Identify the costs associated with decisions and the opportunity costs of alternatives.
- 3.2 Identify the relevant costs and benefits of a decision.
- 3.3 Differentiate between fixed and variable costs and how decisions that change output influence variable costs.
- 3.4 Differentiate between accounting profit and economic profit.
- 3.5 Describe the fixed-cost fallacy of considering irrelevant costs.
- 3.6 Describe the hidden-cost fallacy of ignoring relevant costs.

Chapter 3 – Main Points

- Costs are associated with decisions, *not* activities.
- The **opportunity cost** of an alternative is the profit you give up to pursue it.
- In computing costs and benefits, consider *all* costs and benefits that vary with the consequences of a decision and *only* those costs and benefits that vary with the consequences of the decision. These are the **relevant costs** and **benefits** of a decision.
- **Fixed costs** do not vary with the amount of output. **Variable costs** change as output changes. Decisions that change output will change only variable costs.
- Accounting profit does not necessarily correspond to real or economic profit.
- The **fixed-cost fallacy** or **sunk-cost fallacy** means that you consider irrelevant costs. A common fixed-cost fallacy is to let overhead or depreciation costs influence short-run decisions.
- The **hidden-cost fallacy** occurs when you ignore relevant costs. A common hidden-cost fallacy is to ignore the opportunity cost of capital when making investment or shutdown decisions.
- EVA® is a measure of financial performance that makes visible the hidden cost of capital.
- Rewarding managers for increasing economic profit increases profitability, but evidence suggests that economic performance plans work no better than traditional incentive compensation schemes based on accounting measures.

Warm-Up

- Think of a time you said 'yes' to something: a job, a class, an event and later realized it cost you more than just money. What did you give up to do it?

Benefits, Costs, and Decisions

- In the previous lecture, we talked about benefits and costs
- We extend that analysis by measuring different types of costs
- We need to make sure we understand the costs involved in any decision
- While making our decisions we have to be sure to avoid the sunk-cost fallacy and hidden cost fallacy (more on this later)

Fixed, Variable and Total Costs

- **Fixed costs (FC)** do not vary with the quantity of output produced.
 - For example: lease or rent payments
 - Other examples:
cost of equipment, loan payments, insurance
 - Fixed costs only exist in the short run.
- **Variable costs (VC)** vary with the quantity produced.
 - For example: wages paid to workers
 - Other example: cost of materials
- **Total cost (TC)** = $FC + VC$

Activity – Finding Costs

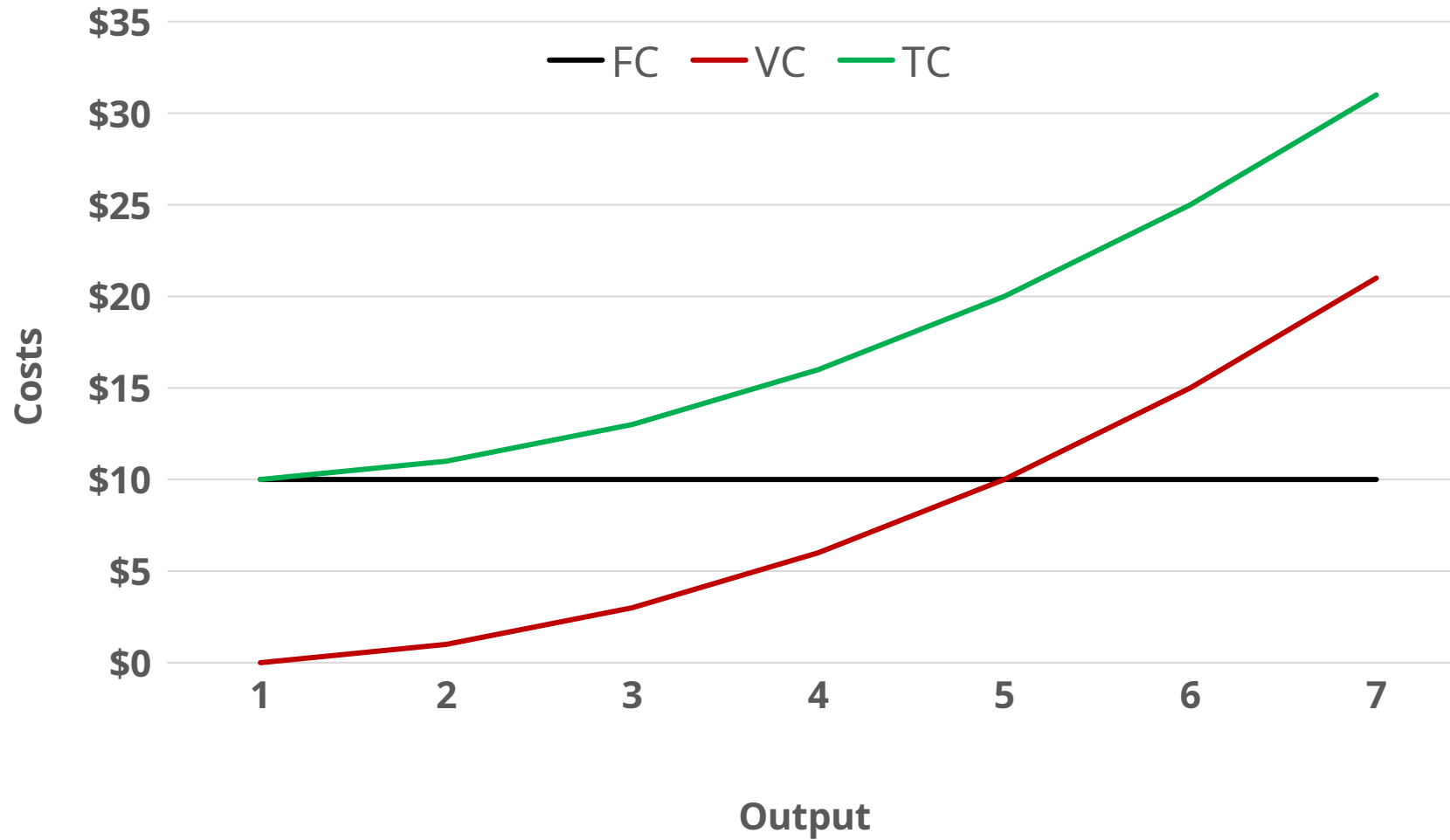
- Determine fixed cost, variable cost and total cost

Quantit y	Variable Costs	Total Costs	Fixed Costs
0			\$10
1	\$1		
2	\$3	\$13	
3	\$6	\$16	
4	\$10		
5		\$25	
6	\$21		\$10

Activity – Finding Costs

Quantity	Variable Costs	Total Costs	Fixed Costs
0	\$0	\$10	\$10
1	\$1	\$11	\$10
2	\$3	\$13	\$10
3	\$6	\$16	\$10
4	\$10	\$20	\$10
5	\$15	\$25	\$10
6	\$21	\$31	\$10

Cost Curves



Another Cost Example

- A business incurs the following costs per unit: Labor at \$100/unit, Materials at \$50/unit, and rent at \$50,000/month. If the firm produces 50,000 units a month, the total variable costs equal

$$\text{Variable Costs} = 50,000(100) + 50,000(50) = \$7,500,000$$

- Keeping with the same scenario: what are the firms total costs?

$$\text{Total Costs} = \text{Fixed Costs} + \text{Variable Costs}$$

$$\$50,000 + \$7,500,000 = \$7,550,000$$

Opportunity Cost

- **Opportunity Cost** – the value of the *next* best (forgone) alternative
- “There ain’t no such thing as a free lunch”
- What’s the opportunity cost of attending class?
- How does opportunity cost hurt you?
 - Consider a piece of unused capital. If the piece of capital were in use it would generate \$5,000 in revenue (output) in a year but depreciate considerably after that. You also have the chance to sell the capital for \$10,000. What should you do? Keep it unused, use it or sell it?
- Example: You have a \$100 concert ticket but choose to study for an exam. The opportunity cost of studying is the value of the concert ticket.

Discussion Activity 1

- Think of a time when you said yes to something but later realized you missed out on a better opportunity. What was the opportunity cost of that decision?
- Consider both time and money. Was the forgone alternative more valuable in hindsight?

Accounting and Economic Costs

- Accounting costs measure explicit costs or costs that can be seen
- **Explicit costs** require an outlay of money, e.g., paying wages to workers.
- Economic costs measure those costs that are explicit and implicit
- **Implicit costs** do not require a cash outlay, e.g., the opportunity cost of the owner's time.

Accounting vs. Economic Profit

You were making \$40,000 a year at a job you didn't like. You now own a business. The rent for land/building is \$10,000 a year. You use \$65,000 in materials and other miscellaneous expenses in your first year. You sell 10,000 shirts at a price of \$15. What are your accounting profits? What are your economic profits?

Accounting Profit = Total revenue – explicit costs

$$(10,000 \times 15) - (65,000 + 10,000) \\ 150,000 - 75,000 = \textbf{\$75,000}$$

Economic Profit = Total revenue – explicit costs + implicit costs

$$(10,000 \times 15) - (65,000 + 10,000 + 40,000) \\ 150,000 - 115,000 = \textbf{\$35,000}$$

Activity: Accounting vs. Economic Profit

- You're thinking of leaving your job (which pays \$90,000/year) to open a café. You expect the following in your first year:
 - Revenue: \$250,000
 - Rent: \$24,000
 - Wages for employees: \$100,000
 - Supplies and other expenses: \$30,000
- What is your accounting profit?
- What is your economic profit?

Activity: Accounting vs. Economic Profit

- **Accounting Profit**

= Revenue – Explicit Costs

= \$250,000 – (\$24,000 + \$100,000 + \$30,000)

= \$96,000

- **Economic Profit**

= Revenue – (Explicit Costs + Implicit Costs)

= \$250,000 – (\$154,000 + \$90,000)

= \$6,000

Discussion Activity 2

- Suppose you're considering leaving a job that pays \$70K/year to start a business. What factors would you include in your economic profit calculation?
- Think beyond the cash outlays—consider your time, lost salary, and any foregone investment opportunities.

Sunk-Cost Fallacy

- **Sunk-Cost Fallacy** – when you consider costs and benefits that do not vary with the consequences of your decision
- Also known as the fixed-cost fallacy
- These costs don't matter because they are irrelevant. You have already incurred them, and you can't recoup them.
 - Anecdote: My mistake in thinking I know a particular driving route
 - Suppose you get free tickets to a game vs. paying for tickets. If a bad storm occurs, are you willing to go the game with free tickets vs. the ones you paid for?

Hidden-Cost Fallacy

- **Hidden-Cost Fallacy** occurs when you ignore relevant costs—those costs that do vary with the consequences of your decisions.
- This fallacy entails you ignoring the opportunity cost of your decisions.
 - *e.g.*, you get offered a 10% raise to move to a big city but ignore the hidden cost of moving there (i.e., higher standard of living of 15%).
 - *e.g.*, you missed out on a profitable investment because you didn't change your convictions about an unprofitable one
 - *e.g.*, a manager keeps employees on the clock, thus increasing hourly costs, when business is slow.

Two Fallacies Side-by-Side

Fallacy	What It Is	Example
Fixed-Cost Fallacy	Considering sunk costs	Going to a bad movie you paid for
Hidden-Cost Fallacy	Ignoring opportunity costs	Delivering product yourself vs. hiring help

Activity – What’s Relevant

- Can you spot the relevant costs?
- For each of the following decisions, decide whether the cost listed is **relevant** or **irrelevant**.

Scenario	Cost
1. You already paid \$500 for a software license, but are switching platforms	The \$500 past payment
2. You’re deciding whether to open on Sundays	Extra \$200 for hourly staff that day
3. You’re launching a new product and need shelf space	\$1,000 to remove current products
4. You already built a website but might pivot business models	\$3,000 spent on old content
5. You’re choosing between two suppliers	Lower shipping cost from Supplier B

Activity – What's Relevant

Scenario	Cost	Relevant?
1. You already paid \$500 for a software license, but are switching platforms	The \$500 past payment	No (sunk)
2. You're deciding whether to open on Sundays	Extra \$200 for hourly staff that day	Yes
3. You're launching a new product and need shelf space	\$1,000 to remove current products	Yes
4. You already built a website but might pivot business models	\$3,000 spent on old content	No
5. You're choosing between two suppliers	Lower shipping cost from Supplier B	Yes

Activity – Fallacy Spotting

- You're the consultant. Spot the fallacy.
- **Scenario 1:**
 - A company refuses to cancel a failing ad campaign because they "already spent \$50K on it."
- **Scenario 2:**
 - A manager reports strong profits but forgets they're using a building that could be rented out for \$200K/year.
- **Scenario 3:**
 - A startup founder keeps running an unprofitable service "because customers love it," even though it's burning resources that could be used elsewhere.

Discussion Activity 3

- Share a time you (or a business) fell for the sunk-cost or hidden-cost fallacy. What would a rational decision-maker have done?
- Did the past investment unduly influence your future choice? Or was a relevant cost ignored?

Cost Concepts: What Matters in Decision-Making

Concept	Definition	Do We Include It?
Fixed Cost	Doesn't vary with output	Ignore in short-run output decisions
Variable Cost	Changes with output	Include
Sunk Cost	Already incurred, unrecoverable	Ignore
Opportunity Cost	Value of next-best alternative	Include
Explicit Cost	Actual cost outlay	Include
Implicit Cost	Foregone benefits (time, investment)	Include in economic profit

Case Study: The Factory that Should Have Closed

A company owns a small manufacturing plant that produces parts for outdated devices. Here's the situation:

- The plant is fully paid off (no rent) but requires \$500,000/year in maintenance and labor to operate.
- It generates \$600,000 in revenue annually.
- The land the plant is on could be sold today for \$1.5 million.
- The company keeps the plant open to “keep people employed” and because “we’ve invested so much into it.”

1. Accounting Profit:

- $\$600,000 - \$500,000 = \$100,000$ profit

2. Economic Profit (with opportunity cost of capital):

- The \$1.5M could be invested at 10% annually = \$150,000 foregone
- Economic Profit = $\$600,000 - \$500,000 - \$150,000 = -\$50,000$ (loss)

3. Fallacies in Play:

- Sunk-Cost Fallacy: “We’ve already invested so much”
- Hidden-Cost Fallacy: Ignoring the cost of not selling the land

If you were the CFO, what would you recommend? Should they close, sell, or continue? How would you explain this to the board?

Economic Value Added (EVA®): Measuring True Profit

- Definition: $EVA^{\circledR} = \text{Net Operating Profit After Taxes (NOPAT)} - \text{Capital Charge}$
 - Capital charge = Value of capital $\times$ Cost of capital
 - Helps measure whether a firm is truly generating value above its cost of capital.
- Why it matters:
 - Helps spot hidden-cost fallacies by including the cost of using capital.
 - Aligns incentives with long-run value creation.
- Simple Example:
 - NOPAT: \$2M
 - Capital used: \$10M
 - Cost of capital: 10%
 - $EVA = \$2M - (\$10M \times 10\%) = \$1M$

Key Takeaways

- Costs are tied to decisions, not just activities, only costs that change with a decision are relevant.
- Opportunity cost is the value of the next-best alternative you give up.
→ Always include it in decision-making, even if there's no cash outlay.
- Fixed costs do not vary with output; variable costs do.
→ Only variable costs matter when deciding how much to produce.
- Accounting profit = revenue – explicit costs
- Economic profit = revenue – (explicit + implicit costs)
- Two common decision-making traps:
 - Sunk-cost fallacy: considering past, unrecoverable costs that shouldn't affect current choices
 - Hidden-cost fallacy: ignoring real, relevant opportunity costs
- Avoid both fallacies by focusing only on future costs/benefits that change based on the decision.
- Economic value is created when decisions maximize economic profit, not just accounting profit.
- Tools like EVA (Economic Value Added) help firms account for opportunity costs in evaluating performance.

Knowledge Check

1. What's the opportunity cost of attending a class instead of working a paid shift?
2. Why shouldn't sunk costs affect future decisions?
3. What's the difference between accounting profit and economic profit?
4. Which of the following is a variable cost: rent, wages, or insurance?
5. What is the hidden-cost fallacy?

Knowledge Check Answers

1. The wages you gave up by not working.
2. They can't be recovered and don't change based on the current decision.
3. Economic profit includes opportunity costs; accounting profit does not.
4. Wages (rent and insurance are fixed).
5. Ignoring relevant costs like opportunity cost or future expenses.

Consultant's Challenge: Cost Clarity

Scenario:

- You're consulting for a small business owner who's debating whether to expand their product line. They're overwhelmed by cost reports and not sure which numbers actually matter.

Your Task:

1. What advice would you give them about which costs are relevant to the decision, and which ones to ignore?
2. How would you help them think clearly about opportunity cost, economic profit, and common fallacies?